

Homework 3

Due on April 19

Important. Only **handwritten solutions** will be accepted; typed solutions are **not** allowed. In all problems, you must show your reasoning and derivations clearly, step by step. Although it is easy to find answers using online tools, the main purpose of the homework is to deepen your understanding through careful, hands-on problem solving. Working through the details yourself is an essential part of learning the material.

1. Let $\iota : S^2 \hookrightarrow \mathbb{R}^3$ be the inclusion of the 2-sphere with unit radius. Let $g = ds^2 = \sum_{i=0}^2 dx^i \otimes dx^i$ be the standard metric of $\mathbb{R}^3$. Find the induced metric ι^*g on S^2 in terms of the polar coordinate of $\mathbb{R}^3$.

$$x^0 = r \sin \theta \cos \phi$$

$$x^1 = r \sin \theta \sin \phi$$

$$x^2 = r \cos \theta$$

Given this metric, find geodesics on S^2 and compute its Riemann, Ricci and scalar curvature.

Do parallel transport of a vector along a triangle ΔPQR on a unit sphere (Figure 1) with respect to the Levi-Civita connection of the metric and find the angle difference when it comes back. Compare it with the area of the triangle (see Homework 1).

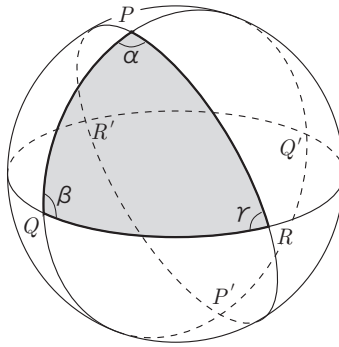


Figure 1: A triangle on a 2-sphere

2. Let $ds^2 = -dt^2 + dx^2 + dy^2$ be the Minkowski metric on $\mathbb{R}^{1,2}$ and $-t^2 + x^2 + y^2 = -1$ for $t > 0$ denote the space-like surface (hyperboloid $\mathbf{S}$). (See Figure 1.) Find the induced metric on the hyperboloid $\mathbf{S}$ in terms of the polar coordinate

$$t = r \cosh \rho$$

$$x = r \sinh \rho \cos \phi$$

$$y = r \sinh \rho \sin \phi$$

Given this metric, find geodesics on $\mathbf{S}$ and compute its Riemann, Ricci and scalar curvature.

3. Let us consider the unit disk $\mathbf{D}$ on the x - y plane. Let $\pi(P)$ be the intersection point of the unit disk and the line between the point $(0,0,-1)$ and $P \in \mathbf{S}$. By assigning $\pi(P)$ to P , there is a one-to-one map between the unit disk $\mathbf{D}$ and the hyperboloid $\mathbf{S}$. Show that the map $\pi^{-1} : \mathbf{D} \rightarrow \mathbf{S}$ is determined by

$$(u, v) \mapsto \left(\frac{2u}{1 - u^2 - v^2}, \frac{2v}{1 - u^2 - v^2}, \frac{1 + u^2 + v^2}{1 - u^2 - v^2} \right).$$

Find the metric on the unit disk pulled back by this map. The unit disk $\mathbf{D}$ with this induced metric is called the Poincaré disk.

The red curve on the hyperboloid is an intersection with a plane (pink in Figure 2,3) that goes through the origin. We put a disk (called Klein's disk) on the bottom of the hyperboloid, which allows us to obtain the corresponding straight line (green) on Klein's disk. This curve is mapped by $\pi : \mathbf{S} \rightarrow \mathbf{D}$ to an (red) arc in the Poincaré disk. If the green line is represented by $x = a$ (Figure 4), find the equation for the red curve.

Find geodesics on $\mathbf{D}$ and compute its Riemann, Ricci and scalar curvature.

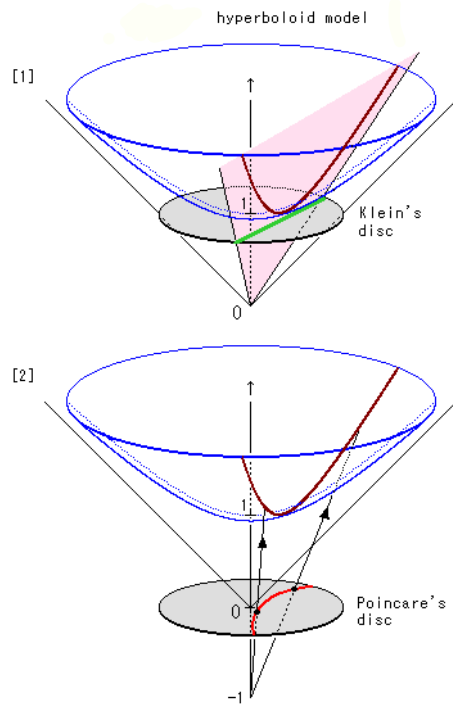
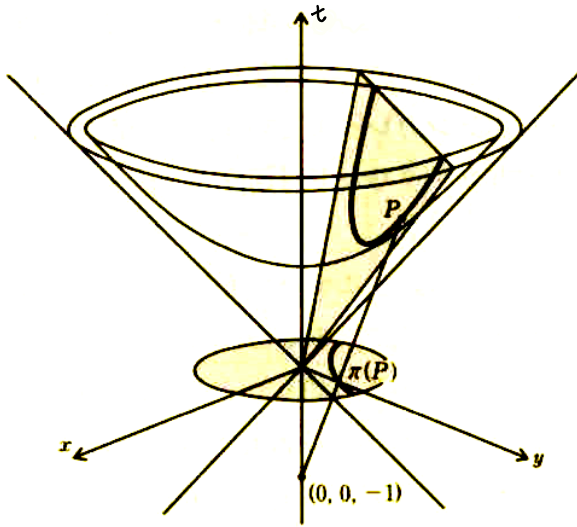


Figure 2: Hyperboloid and Poincaré disk Figure 3: Hyperboloid and Poincaré disk

4. Let $\mathbf{H} = \{(x, y) | y > 0\}$ be the upper half plane (Figure 5). We invert the upper half plane $\mathbf{H}$ to $\mathbf{D}$ in terms of the circle with radius $\sqrt{2}$ around the center $(0, -1)$, and take the reflection with respect to x -axis (Figure 5). This gives a map $J : \mathbf{H} \rightarrow \mathbf{D}; (x, y) \mapsto (u, v)$

$$u = \frac{2x}{x^2 + (y + 1)^2}, \quad v = 1 - \frac{2(y + 1)}{x^2 + (y + 1)^2}.$$

1. Find the induced metric on the upper half plane by this map.

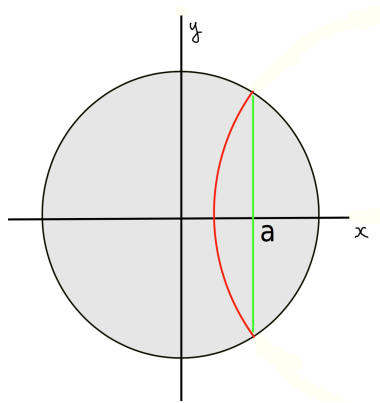


Figure 4: Poincaré disk

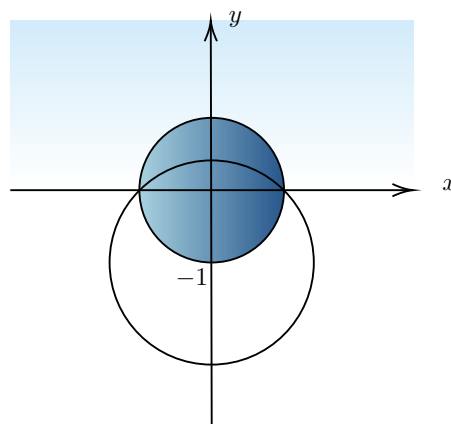


Figure 5: Hyperboloid and Poincaré disk

2. Find geodesics on $\mathbf{H}$ and compute its Riemann, Ricci and scalar curvature.
3. Find the area of the triangle with angles (α, β, γ) bounded by half-circles with respect to the metric (Figure 6). Here, we can use the fact that the area of the triangle on the left of Figure 6 is the same as that of the triangle on the right of Figure 6. Compare with the area of a triangle on the 2-sphere (Problem 1).
4. Do parallel transport of a vector along the triangle with respect to the Levi-Civita connection of the metric and find the angle difference when it comes back. Describe the difference between the sphere and the upper half plane.
5. The **Möbius transformation** of the upper half plane $\mathbf{H} = \{z = x + iy \mid y > 0\}$ is a rational function of the form

$$f(z) = \frac{az + b}{cz + d},$$

where $ad - bc = 1$ with $a, b, c, d \in \mathbb{R}$. If f_1 and f_2 are Möbius transformations, prove that $f_1 \circ f_2$ is also a Möbius transformation. Show that this is an isometry group for the metric.

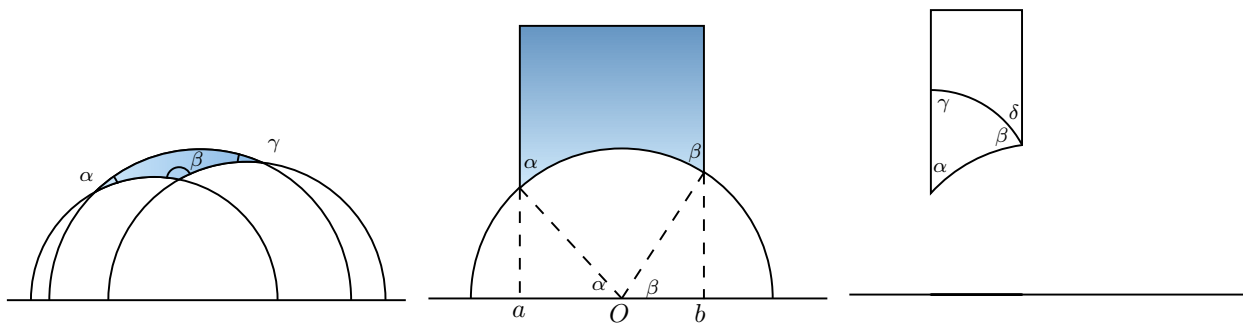


Figure 6: triangle in the upper half plane